

Radiative Transitions of Strange Heavy Axial Mesons in Light-Cone QCD Sum Rules

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Abstract

We study the radiative transitions $D_{s1}(2460) \rightarrow D_s\gamma$, $D_{s1}(2536) \rightarrow D_s\gamma$, and $B_{s1}(5830) \rightarrow B_s\gamma$ with light-cone QCD sum rules. The calculation includes hard photon emission from both quark lines, two-particle photon distribution amplitudes, and three-particle photon distribution-amplitude corrections from the background-gluon term in the quark propagator. The strange-quark mass is retained explicitly. We compare a legacy magnetic-susceptibility photon input with a recent lattice normalization of the leading-twist photon distribution amplitude. For the charm-strange channels we include a physical-current residue normalization calibrated by local QCD sum rules, while for the bottom-strange channel we use a Pullin-Zwicky decay-constant scenario. The preferred lattice-normalized results give a moderate width for $D_{s1}(2460) \rightarrow D_s\gamma$, a strongly cancellation suppressed $D_{s1}(2536) \rightarrow D_s\gamma$ width, and a sub-keV median prediction for $B_{s1}(5830) \rightarrow B_s\gamma$.

1 Introduction

Radiative decays of positive-parity strange heavy mesons provide a sensitive probe of their spin structure, electromagnetic couplings, and proximity to open-flavor thresholds. The charm-strange states $D_{s1}(2460)$ and $D_{s1}(2536)$ have long been discussed in this context, and $D_{s1}(2460) \rightarrow D_s\gamma$ has an experimentally measured branching fraction [1]. The high state $D_{s1}(2536)$ has a measured total width [2], but its radiative decay to $D_s\gamma$ is expected to be small. The bottom-strange state $B_{s1}(5830)$ is observed spectroscopically [3], while its radiative decay remains a target for future searches.

The purpose of this work is to update the light-cone sum-rule analysis of these radiative channels with a transparent separation of hard photon emission, two-particle photon distribution amplitudes (DAs), and three-particle photon DAs. We keep explicit m_s effects and track the normalization associated with the two physical axial currents. We also compare the older magnetic-susceptibility input for the photon DA [4, 5] with the recent lattice normalization of the leading-twist photon DA [6].

2 Conventions and Hadronic Representation

We use the mostly-minus metric, $\epsilon^{0123} = +1$, and the momentum assignment

$$P = p + q, \tag{1}$$

where P is the initial axial-meson momentum, p is the final pseudoscalar momentum, and q is the real-photon momentum. The pseudoscalar interpolating current is

$$J_5(x) = \bar{s}(x) i\gamma_5 Q(x), \tag{2}$$

with $Q = c$ or b . For the axial channel we use the two basis currents

$$J_\mu^A(x) = \bar{s}(x)\gamma_\mu\gamma_5 Q(x), \quad (3)$$

$$J_\mu^B(x) = \frac{i}{m_Q + m_s} \bar{s}(x)\sigma_{\mu\alpha}P^\alpha\gamma_5 Q(x). \quad (4)$$

The tensor current therefore carries the initial-state momentum P^α . The physical-current combinations are taken as

$$J_{1\mu} = \sin\theta J_\mu^A + \cos\theta J_\mu^B, \quad (5)$$

$$J_{2\mu} = \cos\theta J_\mu^A - \sin\theta J_\mu^B, \quad (6)$$

with central $\theta = 35.3^\circ$ and a scan over $25^\circ \leq \theta \leq 45^\circ$.

The transition amplitude is parametrized by an E1 coupling G ,

$$\langle P(p)\gamma(q, \varepsilon) | A(P, \eta) \rangle = e G [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)], \quad (7)$$

so that

$$\Gamma(A \rightarrow P\gamma) = \frac{\alpha_{\text{em}}}{3} G^2 |\vec{q}|^3, \quad |\vec{q}| = \frac{m_A^2 - m_P^2}{2m_A}. \quad (8)$$

3 Light-Cone Sum Rule

The correlation function is

$$\Pi_\mu(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \Gamma_\mu Q(x) \bar{Q}(0) i \gamma_5 s(0) \} | 0 \rangle, \quad (9)$$

where Γ_μ denotes either the axial or tensor basis current. We project onto the transverse structure

$$T_{\mu\nu} = (p \cdot q)g_{\mu\nu} - q_\mu p_\nu. \quad (10)$$

The OPE includes:

- hard photon emission from both the heavy and strange quark lines;
- two-particle photon DAs, including the leading-twist magnetic susceptibility or lattice-normalized transverse photon input;
- three-particle photon DAs generated by the background-gluon insertion in the quark propagator.

Ordinary vacuum-condensate terms are not inserted as separate corrections in the base propagators of the three-point photon LCSR. Condensate dependence enters through photon matrix elements and photon-DA normalizations such as $\chi_s \langle \bar{s}s \rangle$. This is distinct from a local two-point decay-constant sum rule, where ordinary SVZ condensates do appear.

After Borel transformation and continuum subtraction, the basis-current couplings are denoted by G_A and G_B . For the charm-strange physical currents we use

$$G_{2460} = \frac{\sin\theta f_A G_A + \cos\theta f_B G_B}{f_1}, \quad (11)$$

$$G_{2536} = \frac{\cos\theta f_A G_A - \sin\theta f_B G_B}{f_2}. \quad (12)$$

The residue model is calibrated with the working Stage-3 anchor $f_1 = 0.345 \pm 0.017$ GeV; the same two-point overlap model predicts $f_2 \simeq 0.38$ GeV. As a verification, we compare the mixed-current model with the pure axial-current local-QCDSR benchmark $f_A(D_{s1}) = 0.245 \pm$

0.017 GeV from Wang’s Table 1 [7]. In the limiting-current cases the model reproduces $f_1 = f_A$ at $\theta = 90^\circ$ and $f_2 = f_A$ at $\theta = 0^\circ$. At the physical angle, the Wang-like target requires a strongly negative AA/BB overlap and is therefore treated as a diagnostic benchmark, not as a direct determination of the physical mixed residues. For the bottom-strange channel we quote a Pullin–Zwicky decay-constant scenario, $f_{B_{s1}}^{\text{PZ}} = 0.305 \pm 0.027$ GeV and $f_{B_{s1}}^{T,\text{PZ}} = 0.285 \pm 0.027$ GeV [8]. We do not interpret the bottom effective mixed residue as an independently derived local two-point f_1 or f_2 .

4 Numerical Inputs

Table 1 summarizes the main numerical inputs. Masses, quark masses, and electromagnetic normalization are taken from PDG summaries [9]; decay constants and lattice context are taken from FLAG and related heavy-light sum-rule literature [10, 11, 8, 7]. Photon DAs follow Ball, Braun and Kivel [4], the legacy magnetic susceptibility comparison follows Rohrwild [5], and the modern photon normalization scenario uses Bacchio et al. [6].

Quantity	Symbol	Value	Source/use
quark masses	m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV	PDG inputs, with explicit m_s terms
charm hadron masses	$M_{2460}, M_{2536}, m_{D_s}$	2.4595, 2.53511, 1.96835 GeV	PDG
bottom hadron masses	M_{5830}, m_{B_s}	5.82870, 5.36692 GeV	PDG/CMS
lower B_{s1} diagnostic mass	$M_{B_{s1}}^{\text{low}}$	5.750(26) GeV	lattice-inspired diagnostic [12]
D_s and B_s decay constants	f_{D_s}, f_{B_s}	0.2499, 0.2303 GeV	FLAG/lattice context
charm physical residues	f_1, f_2	0.345(17), $\simeq 0.38$ GeV	working Stage-3 anchor and predicted partner
pure axial benchmark	$f_A(D_{s1})$	0.245(17) GeV	local-QCDSR comparison only
bottom PZ constants	$f_{B_{s1}}^{\text{PZ}}, f_{B_{s1}}^{T,\text{PZ}}$	0.305(27), 0.285(27) GeV	PZ scenario
legacy photon input	$\chi_s(1 \text{ GeV})$	+3.15(30) GeV $^{-2}$	comparison scenario
lattice photon input	$f_{\gamma,s}^\perp(1 \text{ GeV})$	−55.1(1.9) MeV	preferred modern scenario

Table 1: Main numerical inputs. The complete bookkeeping tables are `inputs_table.csv`, `outputs/paper_input_summary_table.csv`, and `outputs/input_citation_map.csv`.

5 Results

The final form-factor and width predictions are shown in Table 2. The table quotes medians and 16–84 percentile intervals. Percentile intervals are preferred because several channels, especially $D_{s1}(2536)$ and $B_{s1}(5830)$, are cancellation sensitive and non-Gaussian near zero.

Sector	Decay	Photon input	Mixing	Normalization	G (GeV $^{-1}$)	Γ (keV)
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	legacy χ_s	35.3°	$f_1 = 0.344[0.329, 0.363]$	−0.424[−0.484, −0.360]	37.7 $^{+11.5}_{-10.5}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	legacy χ_s	35.3°	$f_2 = 0.380[0.339, 0.423]$	+0.0118[+0.0014, +0.0233]	0.045 $^{+0.124}_{-0.040}$
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	$f_1 = 0.345[0.330, 0.364]$	−0.277[−0.312, −0.243]	16.1 $^{+4.4}_{-3.7}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	$f_2 = 0.379[0.338, 0.423]$	+0.0403[+0.0194, +0.0596]	0.504 $^{+0.600}_{-0.377}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	35.3°	PZ constants, $f_{\text{eff}} = 0.391[0.354, 0.428]$	−0.0140[−0.0387, +0.0155]	0.109 $^{+0.304}_{-0.101}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	PZ constants, $f_{\text{eff}} = 0.393[0.354, 0.427]$	+0.0356[+0.0029, +0.0694]	0.272 $^{+0.752}_{-0.244}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	25°–45°	PZ constants, $f_{\text{eff}} = 0.394[0.356, 0.430]$	−0.0126[−0.0527, +0.0364]	0.249 $^{+0.570}_{-0.226}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	25°–45°	PZ constants, $f_{\text{eff}} = 0.387[0.348, 0.423]$	+0.0330[−0.0228, +0.0974]	0.429 $^{+1.665}_{-0.401}$

Table 2: Final form-factor and width summary. The lattice photon input is our preferred modern normalization; the legacy susceptibility input is kept for comparison with older photon-DA analyses.

6 Comparison with Experiment

For $D_{s1}(2460)$, BaBar measures a radiative branching fraction $\mathcal{B}(D_{s1}(2460) \rightarrow D_s \gamma) = (16 \pm 4 \pm 3)\%$ [1]. The preferred lattice-normalized partial width, $16.1^{+4.4}_{-3.7}$ keV, corresponds to a total

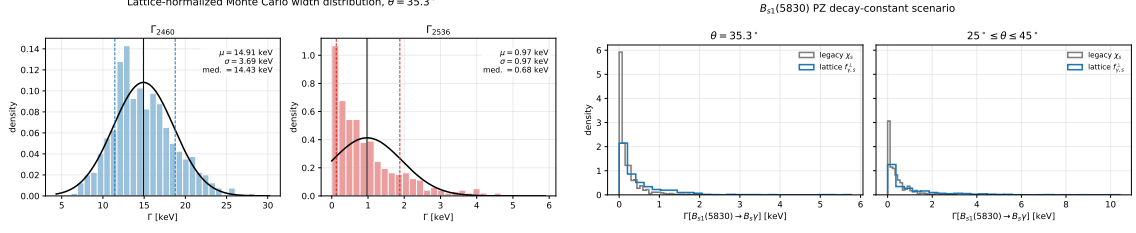


Figure 1: Monte Carlo width distributions for the charm-strange lattice photon-normalization scenario (left) and the bottom-strange PZ decay-constant scenario (right).

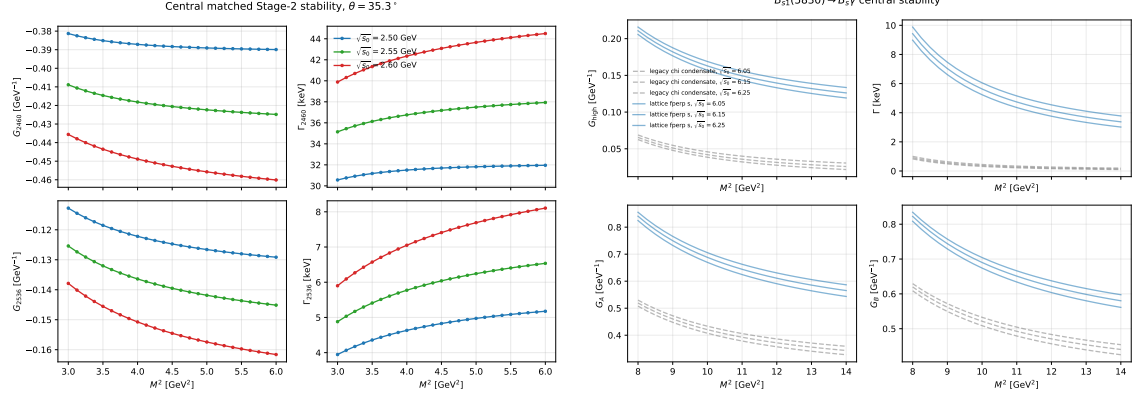


Figure 2: Representative stability plots in the selected Borel and continuum windows for the charm-strange and bottom-strange analyses.

width of order 0.10 MeV if this branching fraction is used literally. This is compatible with the observed narrowness of the state.

For $D_{s1}(2536)$, using the measured total width $\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV [2], the preferred lattice-normalized radiative width gives an estimated branching fraction of order 5×10^{-4} . The channel is therefore strongly cancellation suppressed, which is consistent with the lack of a prominent experimental signal.

For $B_{s1}(5830)$, there is no direct radiative measurement. Our PZ-normalized prediction is a future-search target. The median width remains below 1 keV in both photon-input scenarios, although the mixing-angle scan leaves a long upper tail in the lattice-normalized case.

7 Conclusions

We have constructed a staged light-cone QCD sum-rule analysis of $D_{s1} \rightarrow D_s \gamma$ and $B_{s1} \rightarrow B_s \gamma$ radiative transitions. The calculation includes hard photon emission from both quark lines, two-particle photon DAs, three-particle photon DAs, and explicit strange-quark-mass terms. The modern lattice normalization of the leading-twist photon DA reduces the dominant photon-input ambiguity relative to the older susceptibility-only scenario. The resulting pattern is physically clear: $D_{s1}(2460)$ can have a visible radiative mode while $D_{s1}(2536)$ and the observed high-combination $B_{s1}(5830)$ are cancellation suppressed.

Acknowledgments

To be added.

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